

Agentic Quant Lab

From AI-generated hypotheses to
evidence-backed trading strategies

Agents research.
Deterministic systems decide.



VISION → DISCIPLINED RESEARCH

AI can generate trades. Can we trust them?

TYPICAL AI TRADING BOT

LLM

“Bullish”

BUY

AGENTIC QUANT LAB

*Future only. Today ends at dry run.

Hypothesis

Evidence

Falsification

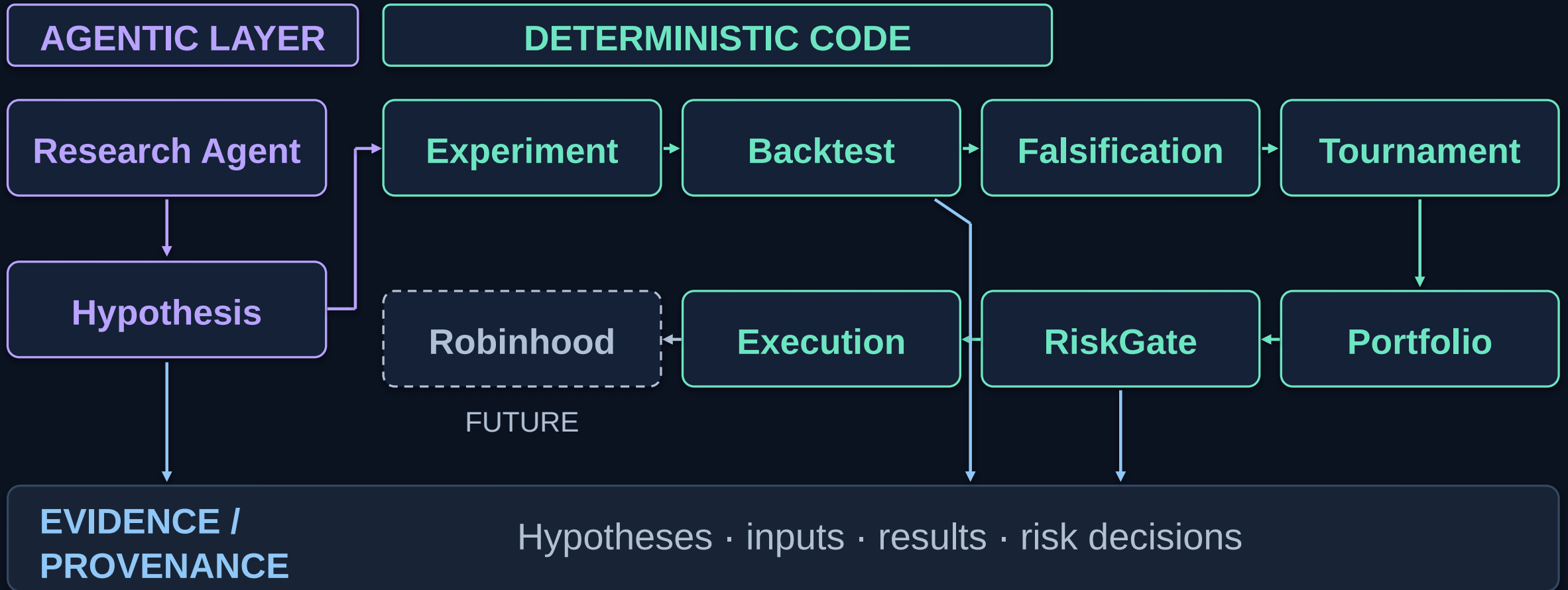
RiskGate

Trade*

Hallucinated reasoning Backtest overfitting Look-ahead / bad data Uncontrolled execution

The hard problem isn't generating strategies.
It's rejecting bad ones.

An autonomous research loop



Target architecture. Scripted agent today; demo receipts ≠ Phase 0 ledger.

The agent has to prove itself wrong

NULL CONTROL

0

No edge?
No alpha.

PLANTED SIGNAL

+

Known edge?
Recover it.

LEAKAGE TRAP

t+1

Future data?
Reject it.

Costs

Frozen specifications

Regime tests

Data provenance

Reject a failing rule. Don't optimize a rescue.

Research requirements, not a claim that the demo validates market alpha.

The system already says “no”

ETF Trend	RESEARCH MORE	Weak vs static de-risking
Stock Momentum	DATA BLOCKED	Universe / identity / delistings
Momentum + Quality	DATA BLOCKED	Stock history + PIT fundamentals
PEAD / Earnings	DEFER	Consensus vintages unavailable
ETF Relative Momentum	NOT YET TESTED	Frozen protocol; acquisition blocked

THIS IS THE POINT. The lab preserves negative evidence.

Trend: near rejection. Missing data is not an economic rejection.

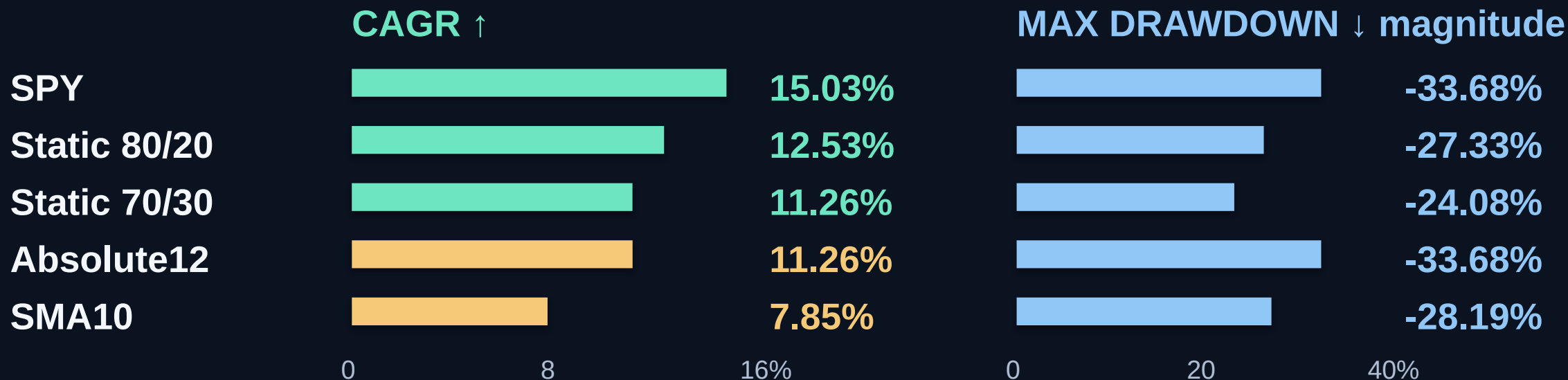
We didn't retune the failed rule.

Trend looked promising — until we tested it properly

REAL RESEARCH · EXPLORATORY / NAV-BASED

NOT PRODUCTION-READY

2016-01-04 → 2026-07-31 · SPY / BIL · 20 bps one-way costs



Absolute12: no worst-drawdown benefit

SMA10: COVID protection, then missed rebound

Losing defensive cycles: Absolute12 4/4 · SMA10 9/11

One command → research to dry-run execution

```
$ uv run aql demo
```

Agent → Experiment → Backtest

Falsification → Tournament → Portfolio

RiskGate → DRY RUN

DATA:
DEMO FIXTURE

LIVE TRADING:
OFF

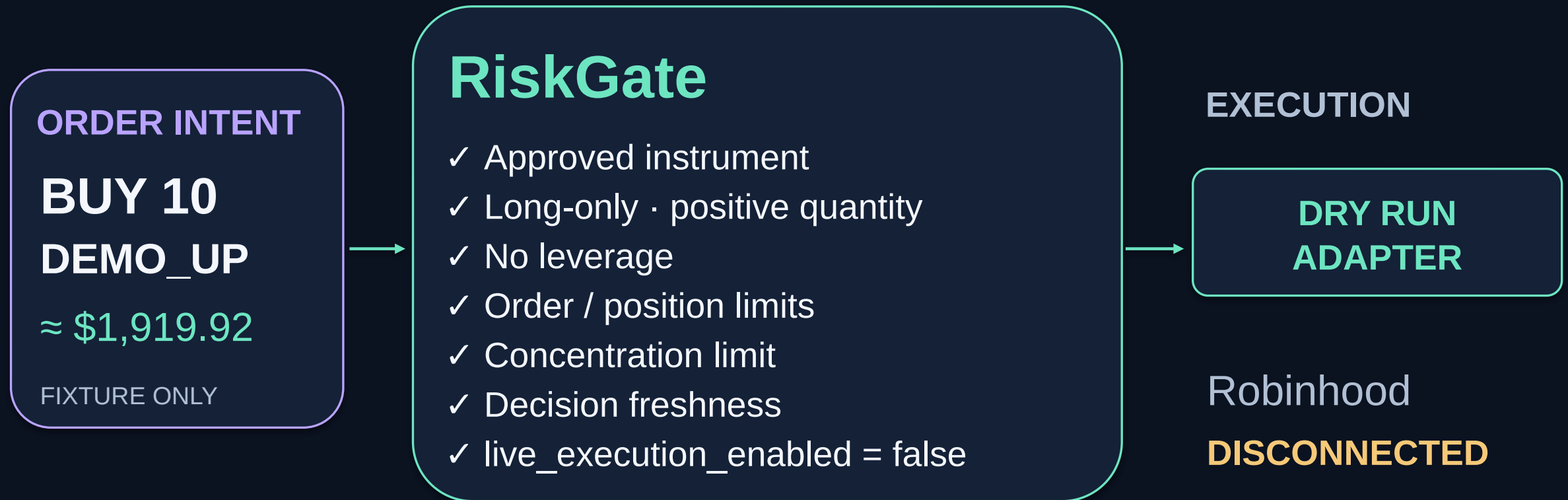
EXECUTION:
DRY RUN

INTERNET:
NOT REQUIRED

Scripted agent. Synthetic results. No LLM or broker call.

Prepared offline: `uv run --offline --no-sync aql demo`

Agents never get direct control of capital



Agents propose. RiskGate authorizes.

One BUY / supplied demo state. No broker reconciliation.

Future design: credentials stay outside research agents.

Where AI actually adds value

AGENT / TARGET CAPABILITIES

- Read research
- Generate hypotheses
- Find failure regimes
- Interpret earnings / guidance
- Propose experiments
- Investigate degradation

DETERMINISTIC CODE

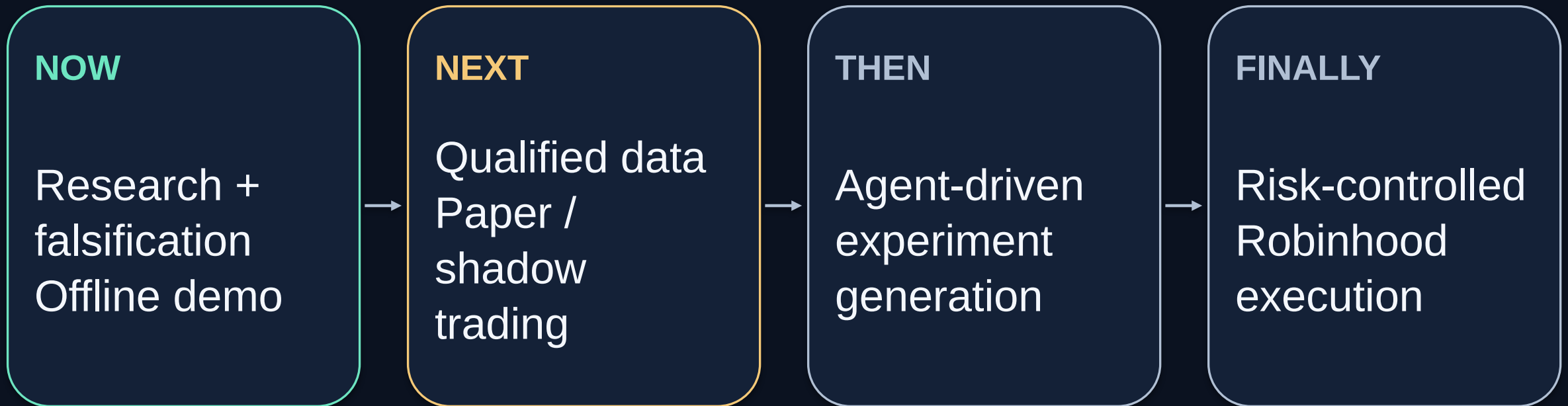
- Calculate returns
- Rank assets
- Run backtests + model costs
- Allocate capital
- Enforce risk limits
- Submit orders (future only)

Use AI for reasoning.

Use code for truth.

“Truth” means reproducible calculations, not certainty about markets.

Roadmap



LIVE TRADING DISABLED TODAY

Promotion requires evidence, not a better story.

The goal isn't an AI that trades more.

**It's an AI that knows
when NOT to trade.**

RESEARCH

EVIDENCE

FALSIFICATION

RISK

Agentic Quant Lab

github.com/BruceGK/agentic-quant-lab